

BOSONIZE-LEAN: Phase 2 Formalization Blueprint

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1 Phase 2 Scope Constraints

Based on the revised Phase 2 directives, this phase formulates the Bosonization Dictionary's density sector. It explicitly excludes Klein factors to prevent context bloat. The mathematical specifications define normal-ordered densities, the exact Schwinger term on bounded energy submodules, the scaled Sugawara Hamiltonian mapping to the free fermion Hamiltonian, and Haldane Completeness via integer partitions.

2 2.1 Density Operators

The raw density is the sum of $c_{q+m}^\dagger c_q$. We normal-order it relative to the Dirac sea to prevent infinite eigenvalues.

```
section Densities
```

```
-- The raw density operator  $\rho(m) = \sum c^\dagger_{q+m} c_q$ . -/
def raw_density (m : LambdaDual L) : Module.End Complex
```

```
  (FockSpace (Chirality x LambdaDual L)) :=
  \sum q : LambdaDual L, (c_mom L (q + m)) * (c_mom L q)
  ^\dagger
```

```
-- Vacuum expectation value of the density mode: \
  \langle 0 | \rho(m) | 0 \rangle -/
```

```
def density_vev (m : LambdaDual L) : Complex :=
```

```

    if m.val = 0 then (L / 2 : Complex) else 0

/-- Normal ordered density mode :\rho(m): -/
def density_mode (m : LambdaDual L) : Module.End Complex
  (FockSpace (Chirality x LambdaDual L)) :=
  raw_density m - LinearMap.smulRight (LinearMap.id) (
    density_vev m)

/-- The zero mode of the normal ordered density is
    exactly the particle number operator N. -/
lemma density_zero_eq_N :
  density_mode (0 : LambdaDual L) = N_op := sorry

end Densities

```

3 2.2 Heisenberg Algebra (Schwinger Term)

The commutator yields the Schwinger term when momenta are equal and opposite ($m + n = 0$), constrained strictly on the budget subspace.

section Schwinger

```

/-- The Exact Schwinger Term evaluated strictly on the
    Budget Subspace. -/
lemma comm_density_modes {K Nmax : Nat} {m n :
  LambdaDual L}
  (h_band : |m.val| + |n.val| + K \le L / 2) (\psi :
    BudgetSpace L K Nmax) :
  (density_mode m * density_mode n - density_mode n *
    density_mode m) \psi.val =
  (if m.val + n.val = 0 then (m.val : Complex) else 0) \
    bullet \psi.val := sorry

end Schwinger

```

4 2.3 Sugawara Construction (Kinetic Energy)

The free fermionic Hamiltonian H_0 is algebraically identical to the bosonic kinetic energy operator H_{sug} on the budget subspace.

`section Sugawara`

```
-- The Free Fermion Hamiltonian (linearized dispersion)
-/
def H_free : Module.End Complex (FockSpace (Chirality x
  LambdaDual L)) :=
  \sum k : LambdaDual L, (k.val : Complex) \bullet (
    c_mom L k * (c_mom L k)^\dagger - LinearMap.
      smulRight (LinearMap.id) (if k.val \le 0 then 1
        else 0))

-- The Bosonic Kinetic Energy Operator (Sugawara
  Hamiltonian) -/
def H_sugawara : Module.End Complex (FockSpace (
  Chirality x LambdaDual L)) :=
  \sum m in (univ.filter (fun (x : LambdaDual L) => x.
    val > 0)),
    density_mode (-m) * density_mode m

-- Sugawara Equivalence -/
lemma sugawara_equivalence {K Nmax : Nat} (h_band : K \
  le L / 4) (\psi : BudgetSpace L K Nmax) :
  H_sugawara \psi.val = (H_free - LinearMap.smulRight (
    LinearMap.id) ((N_op * (N_op + LinearMap.id)) / 2))
    \psi.val := sorry

end Sugawara
```

5 2.4 Haldane Completeness

`section Completeness`

```
-- The cyclic Submodule generated by the negative
  density modes acting on the vacuum state. -/
```

```

def density_cyclic_submodule (K Nmax : Nat) : Submodule
  Complex (FockSpace (Chirality x LambdaDual L)) :=
  Submodule.span Complex (Set.range (fun (p :
    MvPolynomial (Fin K) Complex) =>
    sorry -- Yellow Team Note: Implement the evaluation
      of the multivariate polynomial of density
      operators here.
  ))

/-- Haldane Completeness -/
lemma haldane_completeness {K Nmax : Nat} (h_band : K \
  le L / 4) :
  Module.finrank Complex (density_cyclic_submodule K
    Nmax) = Nat.Partition.partitions K := sorry

end Completeness

```